PRAMANEX LEDGER-Q

Quality, Compliance & AI Assurance OS

Unified Portfolio Specification - 20 Core Features / 12 Shared Layers - Integrated Assurance Edition

DOMAIN SPINE	TRUST SPINE	BOUNDARY
Pharmaceutical Quality + Compliance	AI Governance + Assurance	Human authority remains final

Operating logic: SOURCE -> REQUIREMENT -> GAP/EVENT -> INVESTIGATION -> RISK -> RCA/CAPA/CHANGE -> TRAINING/ACTION -> HUMAN QA DECISION -> EFFECTIVENESS -> AUDIT/INSPECTION EVIDENCE

Parallel AI trust logic: MODEL/AI -> INTENDED USE -> PROVENANCE -> EVALUATION -> GROUNDING -> RISK -> HUMAN OVERRIDE -> RELEASE STATUS -> MONITOR -> CHANGE -> RETEST

Integrated control: material quality changes can invalidate dependent AI assurance; material AI changes can reopen affected SOP, training and quality review. Impact is traced through the evidence graph before release.

1. Final 20 Joined Features - Integrated Quality + AI Assurance

The feature set retains the original 20 non-duplicative capabilities while embedding bidirectional quality-AI invalidation, event forensics, version-bound release status, assurance-debt/blast-radius control and a human-correction firewall inside the features where they operationally belong. No extra top-level feature count is created.

ID	Feature	Primary work	Main layers
F01	Controlled Quality Source & Document Registry	Intake approved SOPs, policies, controlled forms and quality sources; preserve owner, permission, effective date, hash and version history.	L01-L04, L06
F02	Requirement Extraction & Exact Evidence Traceability	Convert source text into structured requirements while every material field remains linked to exact source/version/location.	L04-L07, L09-L11
F03	Regulation-to-SOP / Requirement Mapping	Connect external requirements to applicable SOPs, procedures, processes and versions; retain explicit coverage and applicability rationale.	L04-L07, L09-L11
F04	Compliance Gap, Conflict & Evidence-Sufficiency Control	Detect missing requirements, conflicting evidence, stale coverage, unsupported claims and incomplete evidence. Maintain assurance-debt status for unresolved/stale evidence and surface dependent work that cannot safely progress.	L05-L08, L10-L11
F05	Quality Risk, Severity & Confidence Assessment	Separate quality impact/severity, likelihood, controls, evidence sufficiency and confidence; escalate high-risk or uncertain cases without disguising uncertainty as certainty.	L06, L08, L10-L11
F06	Deviation Investigation Review	Build reconstructable deviation chronology, evidence, investigation candidates, gaps and reviewer disposition. If AI contributes, preserve the exact run, evidence, output and human use; AI may propose hypotheses but never determine root cause or disposition.	L04-L09, L10-L11
F07	OOS/OOT Investigation Review	Review laboratory OOS/OOT chronology, source evidence, conflicts, missing data and human disposition. Preserve AI contribution and reviewer actions so the full event can be reconstructed later.	L04-L09, L10-L11
F08	Bidirectional Change-Control & Impact Assessment	Trace quality changes across requirement, SOP, process, document, product/batch, training, quality records, AI use and dependent outputs. A material model/prompt/tool/data/policy change likewise calculates affected SOPs, training and quality reviews, reopens applicable controls and marks dependent assurance stale until revalidated.	L04, L06-L12
F09	RCA-CAPA Lifecycle & Effectiveness Verification	Evidence-linked root-cause hypotheses -> CAPA candidate -> owner/action -> implementation -> effectiveness check -> human closure. No autonomous root-cause determination, disposition or closure.	L04, L06-L12
F10	Training Impact & Retraining Traceability	When SOP/change/CAPA or a material AI/system change alters controlled work, identify affected roles/training, create retraining or review actions and preserve completion evidence against the correct version.	L02, L04, L06-L08, L10-L12

1. Final 20 Joined Features - Continued

The second half of the catalogue governs oversight, human authority, AI assurance and the continuous release/revalidation loop. Human authority remains the final regulated decision boundary.

ID	Feature	Primary work	Main layers
F11	Audit / Inspection Finding & Readiness Management	Finding -> evidence gap -> action/CAPA -> owner -> response -> closure, while producing a reconstructable inspection-ready evidence chain and exposing open assurance debt.	L02, L04, L06-L08, L10-L12
F12	PQR/APQR & Quality Trend Review	Aggregate quality-event patterns, deviations, OOS/OOT, CAPA, recurring signals and assurance exceptions into reviewable trends without turning correlation into automatic conclusions.	L04-L08, L11-L12
F13	Owner, Deadline, Escalation & Closure Control	Assign accountable owner, acknowledgement, due date, escalation, action state and evidence-backed closure for quality events, assurance debt, revalidation and reopened controls.	L02, L08, L10-L12
F14	Human QA Review, Override & Controlled Correction	Named authorised reviewer can Approve, Revise/Correct, Request Evidence, Reject, Suspend or Escalate; rationale, evidence and residual risk are retained. Human corrections never automatically change model memory, prompts, policy rules, retrieval knowledge or training data; reuse requires a controlled verify -> classify -> approve -> version -> retest path.	L02, L04, L08, L10-L11
F15	Audit/Version Reconstruction, Event Forensics & Controlled Export	Reconstruct historical source -> analysis -> model/configuration -> AI run/evidence/output -> human usage/correction -> final decision. Generate version-bound evidence packages and Assurance Release Passports containing intended use, context, evidence, limitations, approver and lifecycle status.	L02-L04, L07-L12
F16	AI Model / Use / Context & Release Registry	Register model/version, owner, intended task, supported/excluded use, dependencies, accountable reviewer and release state. Track passport lifecycle as VALID, SUSPENDED, EXPIRED, REVOKED or REVALIDATED with explicit trigger and rationale.	L01-L04, L06-L08, L10-L11
F17	AI Provenance, Transparency & Contribution Record	Preserve model, prompt, tools, configuration, assumptions, input dataset/source/version, retrieved evidence, output, authorised use and human interaction for every AI-assisted run so contribution to a regulated event can be reconstructed exactly.	L03-L07, L09-L11
F18	AI Evaluation, Grounding & Hallucination Suite	Frozen expected-vs-actual cases; citation/evidence checking; unsupported-output detection; deterministic-vs-model assessor conflicts remain visible and are routed for accountable review.	L05-L07, L09-L11
F19	Bias, Edge-Case & Consistency Verification	Defined subgroup/edge/adverse cases, repeated-run variation and model/version comparison with denominators, confidence and limitations.	L04, L06, L09-L11
F20	Assurance Monitoring, Regression, Blast Radius & Revalidation	Monitor failures/exceptions/review time/abstention/task success; replay failures; calculate assurance debt and blast radius across sites, products, batches, SOPs, training, CAPAs, prior outputs, workflows and models. Material quality or AI changes invalidate stale assurance and trigger retest; approved corrections may become regression cases but never automatic memory/training updates; release-passport status updates with the outcome.	L04, L06-L12

2. Final 12 Shared Architectural Layers

The 20 capabilities still reuse the same 12 implementation layers. The new assurance behavior is embedded into the appropriate shared controls rather than creating extra layers.

Layer	Layer name	Work inside LEDGER-Q
L01	Workspace & Tenant Isolation	Separate each organisation/workspace and prevent cross-tenant quality, model or evidence leakage.
L02	Identity, Roles & Approval	Authenticate users; enforce QA reviewer, owner, SME, administrator and approver authority.
L03	Authorised Source & Intake	Admit only permitted SOPs, quality records, datasets, model inputs and other approved sources.
L04	Storage, Metadata, Versioning & Release Records	Preserve originals, hashes, metadata, model/prompt/configuration versions, immutable before/after lineage, AI-run records and version-bound Assurance Release Passports.
L05	Domain Parsing & Normalisation	Parse documents, tables and quality records into typed structured records without losing source coordinates.
L06	Source / Policy / Context & Staleness Governance	Apply intended use, source authority, quality policy, jurisdiction/context, AI context-of-use, staleness rules and bidirectional invalidation triggers.
L07	Evidence Graph, Retrieval, Assurance Debt & Blast Radius	Connect regulation, SOP, requirement, deviation, CAPA, training, product/batch, model, decision and prior-output objects; retrieve permission-filtered evidence and calculate affected dependencies when evidence or assurance becomes stale.
L08	Workflow State, Reopening & Orchestration	Control workflow states, human queues, waits, retries, escalation, correction loops, re-opened quality/training reviews, revalidation tasks and safe failure states.
L09	Multi-Agent & Multi-Model Routing	Route bounded tasks to requirement, deviation, OOS/OOT, change, CAPA and AI-assurance specialists/models while preserving task and model boundaries.
L10	Safety, Security & Human Decision	Enforce permissions, validators, no autonomous regulated disposition/root cause/closure, human override, suspension and controlled release boundaries.
L11	Evaluation, Audit, Correction Firewall & Revalidation	Run frozen tests, adverse cases, grounding, bias/consistency tests, regression/retest, passport-state transitions, audit history and KPIs. Human corrections remain governed evidence and cannot self-train or silently rewrite memory/policy.
L12	Integration, Operations & Scale	Support APIs, QMS/RIM integration boundaries, monitoring, notifications, controlled exports, recovery, rollback and later enterprise scale.

Design rule: 20 features do not create 20 layers. All product capabilities reuse the same 12 controlled implementation layers and the same human-release boundary.

3. Simultaneous Feature-Layer Linkage

Workflow stage	Features working together	Architectural layers underneath	Integrated behavior
1. Control the truth	F01 -> F02 -> F03 -> F04	L01-L07	Approved source -> exact requirement -> mapped control -> gap/evidence debt.
2. Understand quality risk	F05 + F06 + F07 + F08	L04-L11	Risk/investigation plus bidirectional change impact and dependency reopening.
3. Correct and close quality problems	F09 + F10 + F13	L02, L04, L06-L12	RCA/CAPA -> training/action -> accountable owner -> effectiveness -> closure.
4. Inspection / quality-system oversight	F11 + F12 + F15	L02, L04, L06-L12	Inspection evidence, trends, event forensics and controlled evidence/passport export.
5. Human authority	F14 governs F05-F20	L02, L08, L10-L11	Approve/revise/reject/suspend; corrections cannot silently self-train the system.
6. Govern the AI itself	F16 -> F17 -> F18 -> F19 -> F20	L01-L12; especially L04/L06/L09-L11	Register -> provenance -> evaluate -> challenge -> monitor/revalidate with lifecycle status.
7. Continuous trust loop	Any material quality/AI change -> F08/F20 -> F14 -> retest -> F15	L06-L12	Invalidate stale assurance, calculate blast radius, reopen impacted controls, then issue updated evidence/passport only after human approval.

4. Integrated Runtime

QUALITY / COMPLIANCE STREAM	SOURCE -> REQUIREMENT -> GAP/EVENT -> INVESTIGATION -> RISK -> RCA/CAPA/CHANGE -> TRAINING/ACTION -> HUMAN QA DECISION -> EFFECTIVENESS -> AUDIT/INSPECTION EVIDENCE
AI ASSURANCE STREAM	MODEL/AI -> INTENDED USE -> PROVENANCE -> EVALUATION -> GROUNDING -> BIAS/CONSISTENCY -> RISK -> HUMAN OVERRIDE -> RELEASE STATUS -> MONITOR -> CHANGE -> RETEST
CONVERGENCE & CHANGE LOOP	Any material change is checked in both directions. The evidence graph calculates what became stale and what is affected; impacted controls reopen; human reviewers decide; approved retest updates the version-bound release status and reconstruction package.

5. Controlled Release Logic

VALID	SUSPENDED	EXPIRED	REVOKED	REVALIDATED
Approved evidence and context remain current.	Temporary hold while an issue or change is investigated.	Time/context/version trigger requires revalidation.	Evidence or control failure makes prior release unusable.	Required retest and human approval restore controlled use.

6. Portfolio Positioning

Project	Core proof
BEACON - Regulatory Change Monitor OS	Regulatory Intelligence + RegOps + change -> impact -> accountable closure
FORGE - CMC Documentation AI	CMC source truth + structured checking + traceable regulatory-document execution
LEDGER-Q - Quality, Compliance & AI Assurance OS	GxP quality + QMS/compliance + CAPA/change + inspection readiness + AI governance/assurance + bidirectional invalidation + reconstructable AI contribution + controlled release status

Portfolio story: Regulatory Control -> CMC Execution -> Quality & Digital Trust

PRAMANEX LEDGER-Q

QualityOS Completion Addendum - Final Integrated Allocation

PRESERVATION RULE

The original five-page Integrated Master Specification is preserved unchanged. This addendum does not replace, renumber, delete, or rewrite F01-F20 or L01-L12. It only allocates the remaining explicit QualityOS source capabilities into the existing LEDGER-Q architecture as child controls and acceptance contracts.

Why this addendum exists

The current master already covers the core QMS workflow: controlled sources, requirement mapping, evidence sufficiency, quality risk, deviations, OOS/OOT, change control, RCA-CAPA, training, inspection readiness, PQR/APQR trends, accountable closure, human QA review, reconstruction, and AI assurance. However, the original QualityOS source concept explicitly names three additional operational capabilities that were not all stated literally in the five-page master: automated production batch auditing trackers, predictive CAPA correction frameworks, and computer-vision labeling-line audits.

Completion decision

QualityOS source capability	Existing LEDGER-Q coverage	Final allocation	Status after addendum
Automated production batch auditing trackers	Batch references exist in F08/F12/F20, but no dedicated batch-audit child workflow.	LQ-Q01 Production Batch Quality Audit	FULL ARCHITECTURAL COVERAGE
Predictive CAPA correction frameworks	Strong RCA-CAPA, trend, risk and effectiveness base in F05/F09/F12/F20.	LQ-Q02 Predictive Quality and CAPA Signal Engine	FULL ARCHITECTURAL COVERAGE
Computer-vision labeling-line audits	AI assurance exists, but the visual labeling-line workflow was not explicit.	LQ-Q03 Labeling-Line Visual Inspection	FULL ARCHITECTURAL COVERAGE

Result

After this allocation, QualityOS can be represented as fully covered at the architecture/specification level inside LEDGER-Q, while human QA authority, evidence traceability, and the no-autonomous-release boundary remain unchanged.

7. QualityOS Child Control LQ-Q01

Production Batch Quality Audit

Purpose

Create a controlled, evidence-linked quality audit of a production batch without converting software output into autonomous batch disposition. The child control turns batch records, specifications, test results, deviations, OOS/OOT records and approved quality rules into reviewable findings.

Allocation into existing features and layers

Binding	Role in LQ-Q01
F01 / F02 / F03	Register approved quality sources, extract structured requirements, and map specifications/SOPs to the batch context.
F05	Assess severity, likelihood, evidence sufficiency and confidence for each batch-quality finding.
F06 / F07	Link batch findings to deviation and OOS/OOT investigation records where applicable.
F08	Trace batch-relevant changes to requirements, SOPs, process parameters, training and quality records.
F12	Aggregate recurring batch signals into PQR/APQR and quality trend review.
F14 / F15	Named QA reviewer decides; evidence and the historical decision are reconstructable and exportable.
L03-L11	Authorised intake, versioning, parsing, context, evidence graph, workflow, bounded specialist routing, human safety gate and audit/evaluation.

Workflow

Approved batch package -> requirement/specification map -> expected-vs-actual checks -> missing/conflicting data checks -> deviation/OOS/OOT linkage -> risk and confidence -> QA review -> quality-audit finding -> trend/PQR linkage -> reconstruction/export.

Minimum output contract

Object	Required fields
batch_audit	batch_id, product_id, site, process_step, source_versions, specification_version, audit_status
batch_finding	parameter, expected, observed, method, unit, evidence_ref, severity, confidence, related_event_id, reviewer_state
decision	reviewer, exact revision, rationale, evidence, residual risk, timestamp

Safety boundary

Allowed: detect, compare, flag, trend and assemble evidence. Not allowed: autonomous batch release/rejection, final OOS disposition, or unsupported root-cause determination.

8. QualityOS Child Control LQ-Q02

Predictive Quality and CAPA Signal Engine

Purpose

Strengthen the existing RCA-CAPA lifecycle by detecting recurring or emerging quality patterns early and creating evidence-backed CAPA review candidates. It is a signal and prioritisation layer, not an autonomous root-cause or CAPA approval engine.

Allocation

Binding	Role in LQ-Q02
F04 / F05	Identify unresolved evidence debt, recurring exceptions and quality-risk patterns; keep uncertainty explicit.
F06 / F07	Use deviation and OOS/OOT histories as controlled event evidence.
F09	Create evidence-linked RCA/CAPA candidates, ownership and effectiveness checks; human closure remains mandatory.
F12	Use PQR/APQR trend review to distinguish repeated patterns from isolated events.
F13 / F14	Assign owner/deadline/escalation and require named QA review before a signal becomes an action.
F20	Monitor recurrence, failures and regression; material changes trigger re-evaluation.
L06-L11	Context, evidence graph, orchestration, bounded model routing, human safety gate and evaluation/audit.

Signal logic

Input evidence	Controlled output
Repeated deviation pattern	RECURRENCE_SIGNAL + supporting event IDs + confidence
OOS/OOT cluster or trend shift	QUALITY_SIGNAL + affected batches/products + reviewer question
Prior CAPA with recurring failure	CAPA_EFFECTIVENESS_REVIEW_REQUIRED
Weak, inconsistent or missing evidence	UNKNOWN / REQUEST_EVIDENCE - never a forced prediction

Acceptance rule

One isolated event must not be promoted into a recurring trend without defined evidence. Predictive language must remain calibrated: candidate signal, not proven cause. Human QA determines the regulated disposition.

9. QualityOS Child Control LQ-Q03

Labeling-Line Visual Inspection

Purpose

Add the explicit computer-vision labeling-line audit capability named in QualityOS while preserving LEDGER-Q evidence and model-governance controls. The visual system flags candidate packaging/labeling exceptions and preserves the exact image/frame, approved reference, model version and human QA disposition.

Supported candidate checks

Visual check	Example controlled result
Correct label/artwork version	MATCH_CANDIDATE / VERSION_MISMATCH
Label present and expected placement	PASS_CANDIDATE / MISSING / MISALIGNED
Print legibility / obvious damage	PASS_CANDIDATE / VISUAL_EXCEPTION / LOW_QUALITY_INPUT
Product-batch-label association	MATCH_CANDIDATE / CONTEXT_MISMATCH / UNKNOWN

Allocation

Binding	Role in LQ-Q03
F01 / F03	Register the approved artwork/label reference and map it to the applicable product, line, batch and SOP.
F05 / F06	Risk-rate a visual exception and, when appropriate, create/link a deviation investigation candidate.
F11	Feed confirmed exceptions and evidence into audit/inspection readiness.
F14 / F15	Human QA reviews the image evidence; historical model/input/output/decision are reconstructable.
F16-F20	Register the visual model/use/context; preserve provenance; test grounding/edge cases; monitor drift and revalidate after model/data/artwork change.
L03-L11	Authorised image intake, versioning, parsing, context, evidence linkage, routing, human decision and evaluation.

Safety rule

Low-quality or ambiguous image evidence must route to NEEDS_REVIEW. A model score may support inspection triage but must not autonomously reject or release a batch.

10. Integrated Feature-Layer Allocation After QualityOS Completion

The three QualityOS child controls reuse the existing LEDGER-Q feature IDs and the same 12 shared layers. They do not create F21-F23 and do not create L13.

Stage	Existing feature path	QualityOS child control	Main layers	Integrated behavior
Batch quality truth	F01-F08, F12, F14-F15	LQ-Q01	L03-L11	Batch package -> specification checks -> findings -> QA review -> trend/reconstruction
Predictive quality action	F04-F09, F12-F14, F20	LQ-Q02	L06-L11	Recurring events -> evidence-backed signal -> CAPA review candidate -> effectiveness monitoring
Visual labeling control	F01, F03, F05-F06, F11, F14-F20	LQ-Q03	L03-L11	Approved artwork + image/frame -> visual exception candidate -> QA decision -> audit/revalidation

Full QualityOS path inside LEDGER-Q

Approved quality evidence -> batch/label context -> requirements and specifications -> automated checks -> quality/visual exception -> deviation or OOS/OOT linkage -> risk -> recurring-pattern signal -> RCA/CAPA candidate -> owner/training/change action -> human QA decision -> effectiveness verification -> PQR/APQR and inspection evidence -> historical reconstruction -> continuous monitoring and revalidation.

Relationship to ComplianceOS and ModelGovernanceOS

QualityOS owns quality-event, batch, labeling, trend and CAPA evidence. **ComplianceOS** supplies SOP/GxP mapping, gaps, audit readiness, training and controlled actions. **ModelGovernanceOS** governs every AI-assisted batch/visual/predictive component through intended use, provenance, evaluation, bias/edge-case testing, monitoring and revalidation. The three remain distinct capability families inside one LEDGER-Q product.

Human authority

F14 remains the regulated decision anchor. None of LQ-Q01-LQ-Q03 changes the rule that root cause, OOS disposition, CAPA approval, batch disposition, release status and final quality decisions require authorised human judgment.

11. Runtime, UI and Agentic Integration Contract

Runtime states

State	Meaning
QUALITY_AUDIT_PENDING	Batch or visual evidence accepted for controlled analysis.
QUALITY_FINDING_CANDIDATE	Evidence-linked candidate exception awaiting QA review.
QUALITY_SIGNAL	Recurring/emerging quality pattern requiring review.
VISUAL_EXCEPTION	Potential label/packaging exception supported by image/frame evidence.
REQUEST_EVIDENCE / UNKNOWN	Evidence insufficient or contradictory; no confident disposition.
HUMAN_QA_REVIEW_REQUIRED	Named QA reviewer must act before controlled progression.
APPROVED / REVISED / REJECTED / SUSPENDED	F14 human decision states with rationale and exact revision.

UI allocation

Screen/workbench	Add without disturbing current product
Quality Command Center	Cards for open batch findings, visual exceptions, recurring signals, overdue CAPA/effectiveness reviews.
Batch Audit Workbench	Specification vs observed table, evidence references, linked deviations/OOS, risk and reviewer state.
Visual Inspection Workbench	Approved reference beside observed image/frame, exception overlay/metadata, model/version and QA decision.
Trend and CAPA Signals	Recurring quality patterns with denominator, evidence, uncertainty and proposed reviewer questions.
Replay / Inspection Evidence	Reconstruct exact batch/image/source/model/output/human decision as it existed at the time.

Bounded specialist routing

Reuse L09 rather than creating a new agent swarm. Route only when needed to: Batch Audit Specialist, Quality Trend/CAPA Signal Specialist, Visual Inspection Specialist, and Evidence/Audit Verifier. All specialist outputs are typed candidates; deterministic specification/version checks run before model interpretation; one controlled write path and F14 human gate remain mandatory.

Data objects

batch_audits, batch_findings, visual_inspections, visual_evidence, quality_signals, capa_signal_links, quality_decisions, quality_effectiveness_checks. Every object carries tenant, source/version, evidence references, workflow revision, reviewer status and timestamps.

12. Acceptance Tests and Final QualityOS Completion Gate

#	Test	Expected result
1	Conforming batch fixture	No unsupported quality finding; evidence references preserved.
2	Out-of-spec or missing batch field	Finding candidate created and routed, not silently normalized.
3	Recurring deviation pattern	QUALITY_SIGNAL with supporting event IDs and denominator.
4	Single isolated event	Not falsely promoted to a trend.
5	Correct approved label image	No false visual exception on the frozen fixture.
6	Wrong/missing/misaligned label fixture	VISUAL_EXCEPTION with exact image/frame evidence.
7	Low-quality/ambiguous image	NEEDS_REVIEW / UNKNOWN, never guessed PASS/FAIL.
8	Human correction	New revision; original output and rationale preserved; checks rerun.
9	Model/specification/artwork version change	Dependent prior assessment becomes stale and revalidation is triggered.
10	Historical replay and restore	Exact evidence, model/configuration, output and human decision reconstruct successfully.

Final architecture status

Constituent capability family	Status in final LEDGER-Q architecture
QualityOS	FULL architectural coverage after LQ-Q01 + LQ-Q02 + LQ-Q03 allocation
ComplianceOS	FULL architectural coverage retained; unchanged by this addendum
ModelGovernanceOS	FULL architectural coverage retained; now also governs QualityOS AI/CV/predictive branches

Claim boundary

This addendum completes the architecture/specification mapping. It does not by itself prove that the batch-audit, predictive-quality or visual-inspection components are coded, tested, deployed, validated for GxP use, customer accepted, or authorised for production release. Promote each child control only through planned -> built -> tested -> demonstrated evidence.

Final locked product statement

LEDGER-Q - Quality, Compliance and AI Assurance OS: QualityOS + ComplianceOS + ModelGovernanceOS fully represented as distinct but integrated capability families across the same 20 canonical features and 12 shared layers, with LQ-Q01-LQ-Q03 completing the explicit QualityOS source scope.